Ranking Transfers, Thresholds Don't: Unsupervised Test-Time Adaptation for Cross-Corpus Audio Deepfake Detection

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***Abstract*—Audio deepfake detectors generalise poorly across corpora despite near-perfect in-domain accuracy. We show why: a fine-tuned self-supervised detector's *ranking* (ROC-AUC) largely transfers cross-corpus even when its decision *threshold* does not calibrate. The model's most-confident predictions on unlabeled target audio therefore remain reliable, enabling stable unsupervised *test-time adaptation* (TTA): we self-train on confident-ranked pseudo-labels and enforce channel-consistency, adapting only LayerNorm and the classifier head, with no target labels and no source data at test time. On a leave-one-corpus-out protocol over four targets (ASVspoof2019, a LibriSpeech-based TTS corpus, In-the-Wild, and a multilingual Arabic set), five seeds each, our method improves EER on three of four targets (In-the-Wild $13.32 \rightarrow 11.23\%$, Arabic $22.07 \rightarrow 20.61\%$, ASVspoof2019 $4.69 \rightarrow 3.36\%$) and is EER-neutral on the fourth. Entropy-minimisation TTA (Tent) is unpredictably catastrophic—stable on four of five ASVspoof2019 seeds, chance-level on the fifth, with no advance warning; a non-SSL backbone collapses on all four targets; a train-time domain-adversarial baseline (DANN) and an aggregation-separation baseline (ASDG-style) both underperform ours on three of four targets despite privileged access to target data during training. A proxy $\mathcal{A}$-distance analysis shows domain separability does *not* decrease after adaptation, ruling out domain-invariance as the mechanism and supporting our ranking-calibration account instead.**

***Index Terms*—audio deepfake detection, anti-spoofing, test-time adaptation, cross-corpus generalisation, self-supervised learning, EER.**

I. Introduction

Self-supervised (SSL) front-ends such as wav2vec 2.0 and XLS-R, paired with lightweight back-ends, reach very low in-domain error on standard anti-spoofing benchmarks [1]–[3]. Deployed on audio from a different source—new microphones, codecs, languages, or generators—accuracy drops steeply, a well-documented generalisation gap [5], [6], [11].

We start from a diagnostic measurement: taking a fine-tuned XLS-R detector and evaluating it on an unseen corpus, its **ROC-AUC stays high while its accuracy at a fixed threshold collapses**—the model still *ranks* clips correctly but its operating point is mis-calibrated for the new domain. Because ranking transfers, the model's most-confident predictions on unlabeled target audio are trustworthy, so we can use them to adapt the model to the target domain *at test time*, without any labels.

Our contributions: (i) the empirical finding that ranking transfers cross-corpus even when the threshold does not, and its consequence for unsupervised adaptation; (ii) a simple, stable TTA method—confident-ranked pseudo-label self-training plus channel-consistency, adapting only LayerNorm and the head; (iii) a four-target, multilingual, five-seed leave-one-corpus-out evaluation against seven comparison points (source-only, Tent, BatchNorm-only recalibration, self-training-only, a non-SSL backbone, DANN, and an ASDG-style aggregation-separation baseline), reported honestly including the one target where the method does not help; (iv) a mechanistic analysis—proxy $\mathcal{A}$-distance before/after adaptation—showing the gain does *not* come from erasing domain identity.

Related work. Frozen or fine-tuned SSL front-ends with AASIST or attentive back-ends dominate recent benchmarks [1]–[4]; we additionally test whether the cross-corpus gap is an SSL question at all via a from-scratch, non-SSL backbone. Train-time domain generalisation (DG) methods align real and separate fake across source domains (aggregation-separation [5]) or use gradient-reversal domain-adversarial training (DANN) [8]; these need target data *during training* rather than at deployment, and we show a much cheaper test-time alternative is not obviously worse. Entropy minimisation (Tent) [7] is the standard TTA recipe in vision but, as we show, is unstable here; batch-statistics recalibration [9] is a no-op for LayerNorm architectures. We borrow the proxy $\mathcal{A}$-distance from Ben-David et al. [10] to test mechanism, and use ASVspoof2019-LA, In-the-Wild [11], and multilingual MLAAD as evaluation corpora.

II. Method

Let x be a waveform with label $y \in \{0,1\}$ (fake=1). A detector f_θ pools frame features into an embedding e and outputs $s_\theta(x) = \sigma(w^\top e + b) = P_\theta(y{=}1 \mid x)$. The encoder is XLS-R-300M with only its top four transformer layers trainable, followed by attentive temporal pooling and a linear head. Its normalisation layers are *LayerNorm*, which carries no running batch statistics—unlike BatchNorm—so a batch-statistics-recalibration baseline is architecturally a no-op here, and confidence-driven gradient adaptation is the minimal mechanism that can move these layers at all.

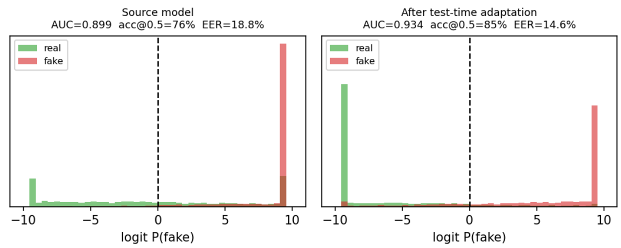

Fig. 1. Detector scores on unseen In-the-Wild audio. **Left:** the source model pushes many real clips past τ=0.5 (accuracy well below AUC) yet still ranks most real below fake. **Right:** TTA shifts the real mass back across the threshold.

The ranking–calibration gap. Ranking quality $\mathrm{AUC} = \Pr[s_\theta(x^+) > s_\theta(x^-)]$ is threshold-free and invariant to reparametrisation of s_θ; accuracy at a fixed threshold τ is not. Empirically, AUC degrades far less than accuracy at τ=0.5 under source→target shift (Fig. 1). We therefore trust score *order*: the extreme quantiles of unlabeled target scores are higher-precision pseudo-labels than a fixed threshold, even when s_θ is mis-calibrated—though this premise weakens when source AUC on the target is itself low (Sec. IV).

Test-time adaptation. Each epoch, we assign confident pseudo-labels from the q- and $(1-q)$-quantiles of target scores, ignoring the middle; self-train with cross-entropy on this confident set ($\mathcal{L}_{\mathrm{ST}}$); and enforce prediction invariance under a stochastic channel perturbation $a(\cdot)$ (gain+noise) via an MSE consistency term $\mathcal{L}_{\mathrm{cons}}$ between clean and perturbed views (stop-gradient on the clean target). The objective is

$$\min_{\{\gamma,\beta\},\,\phi} \; \mathcal{L}_{\mathrm{ST}} + \lambda\,\mathcal{L}_{\mathrm{cons}}, \qquad \lambda{=}0.3,\; q{=}0.3,\; E{=}4, \tag{1}$$

optimising *only* the LayerNorm affine parameters $\{\gamma, \beta\}$ of the fine-tuned layers and the head ϕ; the rest stays frozen. Restricting the trainable set and using only confident tails prevents the degenerate collapse to which entropy minimisation is prone.

Domain divergence. To test whether adaptation works by making source and target indistinguishable in embedding space (as adversarial DG methods explicitly optimise for), we compute a proxy $\mathcal{A}$-distance $\hat{d}_{\mathcal{A}} = 2(1-2\epsilon_h)$ following Ben-David et al. [10], where ϵ_h is a linear domain classifier's held-out error on frozen embeddings, before and after adaptation.

III. Experimental Setup

Protocol. Leave-one-corpus-out over four EER-capable targets: ASVspoof2019-LA, a LibriSpeech-based multi-system-TTS corpus (D2), In-the-Wild [11], and Arabic deepfakes (ArAD). For each held-out target, the source pool merges the other three corpora with a 38-language MLAAD subset (fake-only) for generator/language diversity; held-out target audio, including its MLAAD-overlapping language content, is never in any source pool, and six MLAAD languages are held out of every source pool entirely to test genuine unseen-language generalisation.

Comparison points. Source-only (no adaptation); Tent [7] (entropy minimisation, same trainable set as ours); BN-only (AdaBN-style recalibration [9], no gradient step); self-training-only (ours with λ=0); Oracle (fine-tuned on target labels, a supervised upper bound); RawNet2Lite (a from-scratch, non-SSL backbone in the RawNet2 family [4], trained identically, to isolate whether robustness comes from the SSL front-end); DANN [8] and an ASDG-style aggregation-separation baseline [5] (both trained to convergence with the held-out target's *unlabeled* inputs available during source training—strictly more privileged access than our method has).

Implementation. 16 kHz audio, top-4-layer XLS-R fine-tuning, Adam lr 2×10^{-4}, batch 32, source trained 8 epochs. All results are mean ± std over five seeds (four for Arabic, where a cloud allocation was exhausted before that fold completed).

IV. Results

Table II is the main result. Our method wins on three of four targets: ASVspoof2019 (4.69 → 3.36%, 5/5 seeds), In-the-Wild (13.32 → 11.23%, 4/5), Arabic (22.07 → 20.61%, 4/4); on LibriSpeech-TTS it is EER-neutral (34.44 → 34.61%, improved in only 3/5 seeds), the target with the lowest source AUC (0.72 vs. $\geq$ 0.85 elsewhere)—consistent with our ranking-precondition account: across all 19 (target, seed) points, source AUC and EER gain correlate at r=+0.57 (Fig. 2).

Tent is not merely unstable, it is unpredictably catastrophic. Its ASVspoof2019 per-seed EER is 4.16, 3.91, 3.04, 49.52, 3.42%—four seeds close to source-only, one at chance, with no visible warning sign beforehand. On Arabic it is a *consistent* failure instead (48.16±2.28%, chance-level in all four seeds). Our method shows no comparable collapse on any target (largest std 1.80, on the one target it does not help).

Backbone and prior DG baselines. RawNet2Lite (no SSL pretraining) stays near chance on every target (AUC 0.40–0.65), confirming cross-corpus viability starts from SSL pretraining, not the TTA step; retraining it for 40 epochs with stronger regularisation does not close the gap, ruling out undertraining as the explanation. Both train-time baselines underperform ours on three of four targets despite privileged target access: DANN is markedly worse on Arabic (29.51% vs. our 20.61%) and 3× more expensive (∼30 vs. ∼9 min per target-seed); the least seed-stable method we test (ASVspoof2019 EER ranges 3.23–10.43% across seeds). ASDG beats DANN on three of four targets but beats plain source-only fine-tuning on only one of four (Table I), with a monotonic seed-to-seed drift on In-the-Wild (13.90, 19.78, 28.05%) reminiscent of Tent's instability. Neither train-time baseline is a reliable route to cross-corpus robustness for this task, and a lighter-weight test-time mechanism is not just cheaper but also more effective.

Mechanism. An inductive check (adapt on one half of the target, evaluate on the disjoint half) confirms the gain is not transductive memorisation. On *every* target, the proxy $\mathcal{A}$-distance and a linear domain-origin probe both increase slightly after adaptation rather than decreasing (e.g. Arabic

TABLE I

TEST-TIME ADAPTATION VS. TWO TRAIN-TIME DOMAIN-GENERALISATION BASELINES, MEAN EER % / AUC (DANN, ASDG: 3 SEEDS; SOURCE, OURS: AS TABLE II).

Target	Source	DANN [8]	ASDG [5]	Ours
ASVspoof2019	4.69 / .991	6.61 / .980	4.45 / .990	**3.36** / .995
LibriSpeech-TTS	34.44 / .718	34.81 / .724	34.46 / .721	**34.61** / .718
In-the-Wild	13.32 / .938	**11.95** / .931	20.58 / .861	11.23 / .959
Arabic	22.07 / .857	29.51 / .724	22.19 / .859	**20.61** / .872

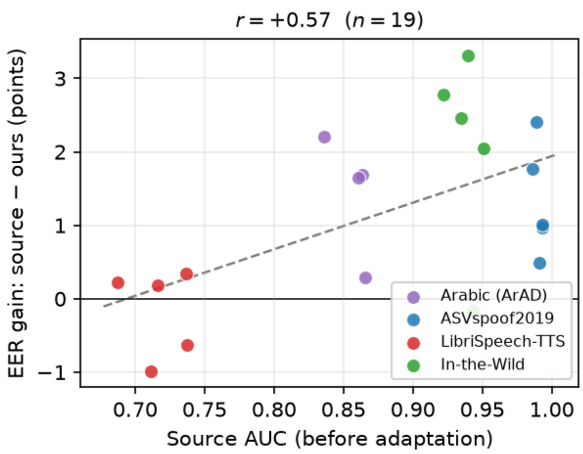

Fig. 2. EER gain (source−ours) vs. source AUC before adaptation, one point per (target, seed), all 19 runs. LibriSpeech-TTS (red), the lowest-AUC target, clusters at or below zero gain; higher-AUC targets cluster above it (r=+0.57).

$\hat{d}_{\mathcal{A}}: 1.545 \rightarrow 1.606$)—the opposite of what domain-invariance would predict, even on targets where EER improves substantially. Adaptation re-shapes the decision boundary using rank-reliable pseudo-labels; it does not erase domain identity, which is independently consistent with why DANN underperforms despite explicitly optimising that quantity. A 38-language MLAAD breakdown shows fake-clip recall is uniform across seen and unseen languages (93.9% vs. 93.3%), so multilingual robustness is not an artefact of language coverage during training.

V. DISCUSSION AND LIMITATIONS

The method is transductive (unlabeled target inputs at adaptation time, the standard TTA setting); an inductive check tracks the transductive numbers closely on every target, including reproducing the LibriSpeech-TTS null result rather than hiding it. We enumerate the remaining limitations explicitly:

- **Needs an already-reliable source ranking.** A from-scratch or low-AUC starting point is not a regime we can currently rescue (Fig. 2).
- **Fixed hyperparameters.** q, λ, E are chosen without target labels and are not per-target optimal; a label-free heuristic we built to fix this failed in the wrong direction (r=+0.74 where r<0 was needed), a negative result consistent with, not contradicting, the ranking-calibration account.

TABLE II

CROSS-CORPUS EER % / AUC, MEAN OVER 5 SEEDS (4 FOR ARABIC; LEAVE-ONE-CORPUS-OUT, TRANSDUCTIVE). BEST OF {SOURCE, TENT, ST-ONLY, OURS} PER COLUMN IN **BOLD**.

Method	ASVspoof2019	LibriSpeech-TTS	In-the-Wild	Arabic
Source-only	4.69 / .991	34.44 / .718	13.32 / .938	22.07 / .857
Tent [7]	12.81±20.52 / .891	34.47±2.36 / .695	26.66±11.09 / .740	48.16±2.28 / .518
Self-training only	3.89 / .989	**34.10** / .704	12.10 / .942	21.85 / .856
Ours (TTA)	**3.36** / .995	34.61 / .718	**11.23** / .959	**20.61** / .872
Oracle (target-supervised)	0.73 / .999	12.89 / .946	3.87 / .993	9.13 / .970

- **Scale.** Four targets and five seeds establish sign consistency, not a rigorous statistical test—though growing from three to five seeds strengthened rather than weakened every central claim, including surfacing Tent's single-seed collapse.
- **SSL dependency.** The gain requires an SSL-pretrained backbone (Sec. IV); it is an adaptation layer on top of an already-transferable representation, not a backbone-agnostic recipe.
- **Single-source scope.** On the harder regime matching the official ASVspoof2021-DF protocol (train on 2019-LA alone), source-only reaches only 26.33% EER and naive TTA actively harms performance under that evaluation set's severe class imbalance (97.5% spoof); we diagnose and partially correct this with a label-free prevalence estimator but do not yet close the gap—a scope boundary, not a retraction of the multi-source results above.

VI. CONCLUSION

Cross-corpus audio deepfake detection fails not because ranking is lost but because the operating point is miscalibrated. A simple unsupervised test-time adaptation exploiting this improves generalisation on three of four leave-one-corpus-out targets, where entropy-minimisation TTA is unpredictably catastrophic, a non-SSL backbone collapses outright, and two independent train-time domain-generalisation baselines underperform it despite privileged data access and 3× the compute. A proxy divergence analysis shows the mechanism is re-calibration of an already class-organised representation, not domain-invariance.

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